

Dear AeroGrid CTO,

The anomaly detector flags two turbines requiring urgent maintenance. **T-04** is failing due to excessive average temperature: **90.6°C**, above the **85.0°C** threshold. **T-07** is failing due to a vibration spike: **25.0mm/s**, above the **15.0mm/s** threshold.

The proposed cloud architecture uses managed ingestion, processing, storage, and monitoring components for a continuous telemetry pipeline. Turbine data should enter through an IoT/event ingestion service, then flow into a streaming layer for buffering and back-pressure control. The containerised Python detector can run in a managed compute service, such as serverless functions or container orchestration, so processing scales automatically as telemetry volume rises. Raw and processed data should be stored in durable object storage, while alerts and metrics are published through monitoring/notification services. This solves the crashing issue by decoupling data ingestion from analysis: if compute workers slow down or restart, the stream retains events instead of overwhelming a single script or server.

For cost optimisation, apply lifecycle policies to telemetry storage: keep recent high-resolution data in hot storage for analysis, then automatically move older raw data to cheaper archive tiers or delete it after the required retention period. This is especially valuable for a massive, continuous pipeline where storage growth is predictable and long-term raw data may be accessed rarely.